

Homework assignments

NYU PHYS-UA 123: Quantum Mechanics I

Fall 2026 Semester

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Submission deadlines: by 11:59pm of due date at top of each homework on Brightspace.

HW1: Units and wave-particle duality (set Sep 3, due Sep 10)

1. Planck's constant $h = 6.626 \times 10^{-34}$ J s, speed of light $c = 2.9979 \times 10^8$ m s⁻¹, and $1 \text{ eV} = 1.602 \times 10^{-19}$ J. Find the numerical value of hc and $\hbar c$ in (a) Joule-metres J m and (b) show that $hc = 1240 \text{ eV nm}$ in atomic units of electronvolt-nanometers eV nm.
2. Sketch a diagram for and describe the setup of the Davisson-Germer experiment.
 - (a) Optional field trip: visit the site where this experiment occurred and find figure 1.
 - (b) How does this experiment show the wave behaviour of electrons? What is meant by wave-particle duality? What is the de Broglie hypothesis and its formula?
 - (c) Derive the expression for the momentum p of an electron with mass m and charge Q accelerated through a potential V . Show that the de Broglie wavelength of an electron accelerated through a voltage V is $\lambda \simeq \frac{1.23 \text{ nm}}{\sqrt{V}}$.
 - (d) What is the numerical value of the voltage required such that the electron de Broglie wavelength is 100 times smaller than (i) a virus with size 20 nm and (ii) the mean diameter of a silicon atom of 0.2 nm?
 - (e) Describe the quantities in the Bragg formula $d \sin \phi = n\lambda$. Dialling the incident electron beam energy of their experiment, Davisson and Germer also found an $n = 1$ peak at $\phi = 44^\circ$ for monocrystalline nickel with lattice spacing $d = 0.215 \text{ nm}$. What is the de Broglie wavelength and energy $E = p^2/2m$ of the incident electron beam? Express your answers in units of eV and nm.
3. Using $E = mc^2$, show that the electron mass $m_e = 9.11 \times 10^{-31}$ kg is 511 keV/c².
 - (a) Convert the following rest masses into eV/c² units:
 - (i) proton $m_p = 1.67 \times 10^{-27}$ kg, (ii) onion cell 10^{-15} kg, (iii) cat $m_{\text{cat}} = 5$ kg.
 - (b) Show that de Broglie wavelength is $\lambda_p = (hc)/(0.01mc^2)$ for a particle with mass m and speed $v = 0.01c$. For the electron and objects (i)–(iii), estimate numerical values for the Compton wavelength λ_C and λ_p for $v = 0.01c$.
4. Describe the Compton experiment and write down the Compton scattering formula.
 - (a) Taking the incident wavelength as $\lambda = 0.01 \text{ nm}$, what are the outgoing scattered wavelengths λ'_θ for scattering angles $\theta = 0^\circ, 90^\circ, 180^\circ$ in the cases of: (i) classical electromagnetism and (ii) treating light as a particle?
 - (b) How does this experiment and its results demonstrate the particle nature of light?

HW2: Atoms and wavefunctions (set Sep 10, due Sep 17)

1. Describe the Frank-Hertz experiment and how the results demonstrate quantized energy levels of atoms. Consider the Bohr model of hydrogen:
 - (a) What is the empirical evidence for the existence of an atomic nucleus?
 - (b) By equating centripetal and electrostatic forces then using Bohr quantization $L = n\hbar$ of angular momentum $L = mvr$, show that the possible radii are $r_n = (\hbar n^2)/(mc\alpha)$ and find the expression for the dimensionless constant α .
 - (c) Find an expression for the electron velocity v with circular Bohr orbits in terms of the speed of light c and quantum number n . Evaluate v/c for $n = 1$.
 - (d) Evaluate the numerical pre-factors to show the energies are $E_n = -13.6 \text{ eV}/n^2$. What is the energy and wavelength of the photon for $n = 3$ transitioning to $n = 2$?
2. State what a wavefunction $\Psi(x)$ physically represents.
 - (a) Using the normalization condition, what are the S. I. units of the wavefunction $\Psi(x)$ as a function of one-dimensional position x ? Explain how you arrived at this answer. How does this change in three spatial dimensions?
 - (b) Given a wavefunction $\Psi(x)$, what is the expression for the probability of finding a particle in $a < x < b$? What mathematical properties must $\Psi(x)$ satisfy to be considered a probability amplitude in quantum mechanics?
 - (c) Given a wavefunction at $t = 0$ is $\Psi(x) = \frac{1}{\sqrt{2\pi}}e^{-(x-a)^2}$, where $a = 1 \text{ m}$, write down the integral representing the probability $P(x)$ of finding a particle between position $x = 0.3 \text{ m}$ and $x = 1.2 \text{ m}$ and sketch the area that is represented by this integral. You do not need to calculate the integral.
3. The double-slit experiment has a source, barrier with slits of width a , and screen.
 - (a) Qualitatively sketch and describe the distribution on the screen for: (i) classical marbles, (ii) classical water waves with wavelength $\lambda \approx a$.
 - (b) Repeat these sketches for N quantum electrons with de Broglie wavelength $\lambda \approx a$ for (iii) $N = 1$ and (iv) $N \simeq 30$.
 - (c) Discuss qualitatively the similarities and differences between the classical and quantum cases. Why are classical probabilities insufficient for quantum physics?

HW3: Schrödinger equation (set Sep 17, due Sep 24)

- Write down the time-dependent Schrödinger equation (TDSE) and explain each term.
 - Why is the TDSE first order in time while Newton's law $f = m\ddot{x}$ is second order?
 - What does the Hamiltonian $\hat{H}$ represent physically? Show that the wavefunction $\Psi(x, t) = Ae^{-i(\omega t - kx)}$ satisfies the TDSE. What is the physical significance of ω and k , $\hat{H}$, and how they are related to differential operators?
 - Explain the condition(s) required and show the derivation from the TDSE to reach the time-independent counterpart. What meant physically by *stationary state*?
 - Integrate the TDSE to derive how an initial wavefunction $\phi(t = 0) = \phi_0$ evolves with time. For a wavefunction initially in state $\phi(t = 0) = e^{i\pi/2}$ with energy E , what is the wavefunction at time $t = T$?
- By taking the time derivative of the probability density $\rho = \Phi^*\Phi$ for the wavefunction $\Phi(z, t)$ and applying Schrödinger's equation, show that it satisfies

$$\frac{\partial \rho}{\partial t} + \frac{\partial j}{\partial z} = 0. \quad (.22)$$

Find the expression for $j(z, t)$. What is the physical significance of this equation? Evaluate $j(z)$ for the plane wave states $\psi = e^{\pm ikz}$ and how is this related to velocity?

- A free particle moves on a circle with radius R obeying Schrödinger's equation with $V(x) = 0$ and periodic boundary conditions $\psi(x) = \psi(x + 2\pi R)$. Show that the momentum p_n is integer n quantized and derive an expression for the energy E_n .
- Write down the Schrödinger equation for wavefunction $\psi(x)$ in the potential

$$V(x) = \begin{cases} 0, & |x| < \frac{a}{2}, \\ \infty & \text{otherwise.} \end{cases} \quad (.23)$$

- What is the boundary condition at $x = \pm a/2$? Show that the energy eigenvalues E_n are integer n quantized and derive the wavefunctions $\psi_n(x)$ for $n = 1, 2, 3$.
- Determine the normalization constant for the wavefunctions in the ground $\psi_1(x)$, first $\psi_2(x)$ and second $\psi_3(x)$ excited states.
- Sketch the forms of the wavefunction $\psi_n(x)$ and probability distributions $|\psi_n(x)|^2$.
- Show that the wavefunctions $\psi_1(x)$ and $\psi_2(x)$ are orthogonal

$$\int_{-a/2}^{a/2} \psi_1(x) \psi_2(x) dx = 0. \quad (.24)$$

HW4: Bound states and wavepacket (set Sep 24, due Oct 1)

1. Consider the finite-well potential in section 6.4 with potential

$$V(x) = \begin{cases} -V_0, & |x| < a, \\ 0, & |x| > a. \end{cases} \quad (.25)$$

- (a) Explain the concept of parity, why the Hamiltonian with the above $V(x)$ has parity symmetry, and what this implies for wavefunction $\psi(x)$ solutions.
- (b) Consider bound states such that $-V_0 < E < 0$. Write down the time-independent Schrödinger equation for the three regions: $x < -a$, $-a < x < a$, $x > a$.
- (c) What functions (e.g. exponential, trigonometric functions) do the odd-parity wavefunction solutions take in the three regions $x < -a$, $-a < x < a$, $x > a$? What expressions do the kinetic energy take in each region? Sketch the even and odd parity wavefunctions with the lowest energy states in relation to the potential (.25).
- (d) Show the steps to derive the condition that odd-parity wavefunctions must satisfy:

$$k \cot(ka) = -\kappa. \quad (.26)$$

State the expressions for k and κ . Find the quadratic relation between k , κ , V_0 by eliminating E from Schrödinger's equation. Sketch this quadratic relation and $k \cot(ka) = -\kappa$, then explain qualitatively how to find the solutions graphically.

2. Consider a particle wavefunction $\psi(x)$ taking the form of a Gaussian wavepacket

$$\psi(x) = \frac{1}{(2\pi\sigma^2)^{1/4}} \exp\left(-\frac{x^2}{4\sigma^2}\right). \quad (.27)$$

- (a) What is the wavefunction normalization condition? Show that this wavefunction is normalized with the help of the standard Gaussian integral identity

$$\int_{-\infty}^{\infty} dx e^{-(a^2x^2+bx)} = \frac{\sqrt{\pi}}{a} e^{b^2/4a^2}. \quad (.28)$$

- (b) Explain qualitatively how the Fourier transform from position $\psi(x)$ to momentum $\psi(p)$ space, $\psi(p) = \frac{1}{\sqrt{h}} \int dx e^{-ipx/\hbar} \psi(x)$, is analogous to a discrete Fourier series. What is the physical interpretation in terms of quantum superposition?
- (c) Perform the Fourier transform integral to show that the wavefunction $\psi(p)$ and its probability distribution $|\psi(p)|^2$ are also Gaussians.
- (d) How are the standard deviations of the wavepacket in position and momentum spaces related mathematically? Comment on its physical significance.

HW5: Scattering and Dirac notation (set Oct 1, due Oct 8)

1. Consider the scattering problem through a thick-wall potential

$$V(x) = \begin{cases} V_0, & 0 < x < a, \\ 0, & \text{otherwise.} \end{cases} \quad (.29)$$

- (a) Write down the Schrödinger equation and the functional forms of the wavefunctions in the three regions $x < 0, 0 < x < a, x > a$ for (i) $E < V_0$ and (ii) $E > V_0$.
- (b) For $E > V_0$, apply appropriate boundary conditions and the definitions of probability currents j to derive the transmittivity $T = |j_{\text{transmitted}}/j_{\text{incident}}|$

$$T = \frac{1}{1 + \frac{1}{4} \left(\frac{k}{K} - \frac{K}{k} \right)^2 \sin^2(Ka)}. \quad (.30)$$

- (c) What is the condition on K for $T = 1$ and thus reflectivity $R = 0$ to vanish?
2. A vector space V has position vectors $\mathbf{r} = (x, y, z)^T$. How do we write $\mathbf{r}$ in terms of basis vectors. How do we write a scalar product with another vector $\mathbf{r} \cdot \mathbf{r}'$ in Dirac notation? What is meant by a Hilbert space and adjoint space?
 3. Given a state $|\psi\rangle = e^{i\pi/6}|a\rangle + (i/2)|b\rangle$, what is its dual $\langle\psi|$?
 4. How do we expand a state $|\psi\rangle$ with basis states $|n\rangle$ and coefficients a_n ? In quantum mechanics, what property must $\langle\psi|\psi\rangle$ satisfy and what is the significance of a_n ?
 5. How do we mathematically define the inner product $\langle\psi|\phi\rangle$ for finite-dimensional vectors and infinite-dimensional functions?
 6. Like the position representation, we can write a state $|\psi\rangle$ in momentum representation

$$|\psi\rangle = \int_{-\infty}^{\infty} dp \psi(p) |p\rangle. \quad (.31)$$

- (a) Explain each object $|\psi\rangle$, $\int_{-\infty}^{\infty} dp$, $\psi(p)$ and $|p\rangle$ in the language of probability amplitudes and basis vectors of a Hilbert space.
- (b) How do we extract the wavefunction in momentum space $\psi(p)$ from state $|\psi\rangle$?
- (c) How do we write the orthogonality condition for the momentum basis states $|p_1\rangle$ and $|p_2\rangle$?

HW6: Operators and measurement (set Oct 8, due Oct 15)

1. What is the definition of a *Hermitian operator* and why is this important for physics? If $\hat{X}$ and $\hat{P}$ are Hermitian operators, show that $i[\hat{X}, \hat{P}]$ is also Hermitian.
2. What is the condition for an operator $\hat{Q}$ acting on wavefunctions $\psi(x), \phi(x)$ to be Hermitian in Dirac and integral notation? Why do $\psi(x)$ and $\phi(x)$ vanish as $x \rightarrow \infty$? Prove via integration by parts that the momentum operator $\hat{p} = -i\hbar \frac{\partial}{\partial x}$ is Hermitian.
3. Consider the state $|\psi\rangle = a|A\rangle + b|B\rangle$.

(a) What is the amplitude $\langle A|\psi\rangle$ and explain your answer? Upon measurement, what is the probability that the state is in $|A\rangle$ and $|B\rangle$?

(b) What is the probability of measuring state $|A\rangle$ if (i) $a = e^{i\pi}$, (ii) $b = i/2$?

4. Given the adjoint of an operator product is $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$, show that $(\hat{A}\hat{B}\hat{C})^\dagger = \hat{C}^\dagger\hat{B}^\dagger\hat{A}^\dagger$. By explicitly writing out the left and right hand sides, prove the following identities

$$[\hat{A}, \hat{B} + \hat{C}] = [\hat{A}, \hat{B}] + [\hat{A}, \hat{C}], \quad (.32)$$

$$[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}. \quad (.33)$$

5. Consider two observables of Hermitian operators $\hat{A}$ and $\hat{B}$ with eigenvalue equations

$$\hat{A}|a_n\rangle = a_n|a_n\rangle, \quad \hat{B}|b_n\rangle = b_n|b_n\rangle. \quad (.34)$$

- (a) Given $\hat{A}$ is Hermitian, show that its (i) eigenvalues must be real, (ii) eigenstates are mutually orthogonal. If the $[\hat{A}, \hat{B}] = 0$ commute, show that they are simultaneous eigenstates of both $\hat{A}$ and $\hat{B}$. Why are these results important for physics?
 - (b) Express the state $|\psi\rangle$ as a expansion of eigenstates $|a_n\rangle$ weighted by amplitudes $\gamma_n = \langle a_n|\psi\rangle$. Using this, derive the expectation value of $\hat{A}$ when acting on $|\psi\rangle$.
 - (c) Consider $[\hat{A}, \hat{B}] \neq 0$ on state $|\psi\rangle$. Let us measure one observable first, then the other immediately after without resetting the system. Show that the probability of measuring $\hat{A}$ first then $\hat{B}$ is $P = |\langle b_n|a_n\rangle|^2 |\langle a_n|\psi\rangle|^2$. Find the probability of measuring $\hat{B}$ first then $\hat{A}$ and show that it differs. Discuss the physical significance.
6. By writing out the differential operators and acting on a general wavefunction $\psi(\mathbf{x})$, prove the canonical commutation relation for any number of dimensions $i, j = 1, 2, \dots, N$

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}, \quad (.35)$$

where the Kronecker delta is $\delta_{ij} = 0$ if $i \neq j$ and $\delta_{ij} = 1$ if $i = j$. What is the physical significance of this commutator for $i = j$ and $i \neq j$? Why does $[\hat{x}_i, \hat{x}_j] = [\hat{p}_i, \hat{p}_j] = 0$?

HW7: Harmonic oscillator and conservation (set Oct 22, due Oct 29)

1. The harmonic oscillator Hamiltonian is $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$ and ladder operators are

$$\hat{A} = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} + i\hat{p}), \quad \hat{A}^\dagger = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} - i\hat{p}). \quad (.36)$$

- (a) Taking the product $\hat{A}\hat{A}^\dagger$, show that the $\hat{H}$ can be written as $\hat{H} = (\hat{A}\hat{A}^\dagger - \frac{1}{2})\hbar\omega$.
- (b) Show that the commutator $\hat{H}$ with the lowering operator is $[\hat{H}, \hat{A}] = -\hbar\omega\hat{A}$.
- (c) Use the previous result to show that $\hat{A}$ acting on $|E_n\rangle$ yields the eigenvalue equation $\hat{H}(\hat{A}|E_n\rangle) = (E_n - \hbar\omega)(\hat{A}|E_n\rangle)$. Comment on the physical significance.
- (d) Consider the operator $\hat{N} = \hat{A}^\dagger\hat{A}$ with $\hat{N}|n\rangle = n|n\rangle$ so we can label energy states by integers n e.g. $|n\rangle = |E_n\rangle$, $|n \pm 1\rangle = |E_n \pm \hbar\omega\rangle$. By taking the modulus squares $|\hat{A}|n\rangle|^2 = \langle n|\hat{A}^\dagger\hat{A}|n\rangle$ and $[\hat{A}, \hat{A}^\dagger] = 1$, show that $\alpha = \sqrt{n}$ in the eigenvalue equation

$$\hat{A}|n\rangle = \alpha|n-1\rangle. \quad (.37)$$

- (e) Derive the ground state wavefunction $\langle x|E_0\rangle$ by solving a differential equation, and show that the excited state $\psi_n(x) = \langle x|E_n\rangle$ is $\langle x|E_1\rangle = Axe^{-x^2/4\ell^2}$, What is the constant A ? Explain why the wavefunctions have well-defined parity.
2. For two non-commuting observables $[\hat{A}, \hat{B}] \neq 0$, what inequality must the product of their uncertainties satisfy? What is the physical significance of this? What is this inequality when considering position $\hat{x}$ and momentum $\hat{p}$? Explain physically why the zero-point energy of the harmonic oscillator is non-zero.
3. Consider an operator $\hat{Q}$ that is time dependent $\partial\hat{Q}/\partial t \neq 0$.

- (a) Taking the time derivative of its expectation value $\langle\hat{Q}\rangle = \langle\psi|\hat{Q}|\psi\rangle$ and using the Schrödinger equation, present the derivation for Ehrenfest's theorem

$$\frac{d\langle\hat{Q}\rangle}{dt} = \frac{i}{\hbar}\langle\psi|[\hat{H}, \hat{Q}]|\psi\rangle + \left\langle\frac{\partial\hat{Q}}{\partial t}\right\rangle. \quad (.38)$$

- (b) If an operator $\hat{A}$ is time independent $\partial\hat{A}/\partial t = 0$ and commutes with the Hamiltonian, what is the physical significance of this equation?
- (c) If the system is in an energy eigenstate $|E\rangle$ such that $\hat{H}|E\rangle = E|E\rangle$, show that $\langle\hat{A}\rangle$ is conserved even if it does not commute with the Hamiltonian $[\hat{A}, \hat{H}] \neq 0$.
- (d) For $\hat{Q} = \hat{H}$, what physically is conserved?

HW8: Angular momentum (set Oct 29, due Nov 5)

1. The projection of the orbital angular momentum operator along the x axis is

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y. \quad (.39)$$

- (a) By using the operator identity $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$ and canonical commutation relation $[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$, show that $\hat{L}_x$ is Hermitian. Why does this imply that $\hat{L}^2$ is Hermitian?
- (b) By writing $[\hat{L}_x, \hat{L}_y]$ as $[\hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \hat{z}\hat{p}_x - \hat{x}\hat{p}_z]$, prove the commutator

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z. \quad (.40)$$

Discuss its physical significance. Hint: the $i\hbar$ arises from the commutator $[\hat{z}, \hat{p}_z]$.

- (c) By explicitly expanding the left- and right-hand side, prove the commutator identity $[\hat{A}^2, \hat{B}] = \hat{A}[\hat{A}, \hat{B}] + [\hat{A}, \hat{B}]\hat{A}$. By writing out the definition of $\hat{L}^2$, use this to prove that $[\hat{L}^2, \hat{L}_y] = 0$. What is the physical significance of this?

2. Write down the definition of the angular momentum lowering operator $\hat{L}_-$.

- (a) Prove that $\hat{L}_-$ satisfies the commutator $[\hat{L}_z, \hat{L}_-] = -\hbar\hat{L}_-$.
- (b) Why does the lowering operator commute with the $\hat{L}^2$ i.e. $[\hat{L}^2, \hat{L}_-] = 0$?
- (c) Show that $|\psi\rangle = \hat{L}_-|\lambda, m\rangle$ is an eigenstate of $\hat{L}_z$ and find its eigenvalue.
- (d) Write down the eigenvalue equations for $\hat{L}^2$ and $\hat{L}_z$. Show that

$$\hat{L}_-^\dagger \hat{L}_- = \hat{L}^2 - \hat{L}_z^2 + \hbar\hat{L}_z. \quad (.41)$$

Use this to show the eigenvalue equation for the lowering operator is

$$\hat{L}_-|\lambda, m\rangle = \alpha_-|\lambda, m-1\rangle. \quad (.42)$$

Find the expression for α_- . Comment on the physical significance of this result.

3. Consider spherical coordinates $x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$. Apply the chain rule to show that the $\hat{L}_z$ operator in spherical coordinates is

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}. \quad (.43)$$

HW9: Spherical Schrödinger problem (set Nov 5, due Nov 12)

1. The total angular momentum operator in spherical coordinates (r, θ, ϕ) is

$$\hat{L}^2 = -\hbar^2 \left[\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]. \quad (.44)$$

Why do $\hat{L}^2$, $\hat{L}_z$, and the Hamiltonian $\hat{H}$ commute for central potentials $V = V(r)$? Explain the physical significance.

2. Write down the eigenvalue equations for $\hat{L}^2$ and $\hat{L}_z$. Show that the following spherical harmonics are (i) eigenstates of $\hat{L}^2$ and $\hat{L}_z$, (ii) orthogonal by integration:

$$Y_1^0 = \sqrt{\frac{3}{4\pi}} \cos \theta, \quad Y_1^1 = -\sqrt{\frac{3}{8\pi}} e^{+i\phi} \sin \theta, \quad Y_1^{-1} = \sqrt{\frac{3}{8\pi}} e^{-i\phi} \sin \theta. \quad (.45)$$

3. The time-independent Schrödinger equation in spherical coordinates is

$$\left[-\frac{\hbar^2}{2m_e} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} \right) + \frac{\hat{L}^2}{2m_e r^2} + V(r) \right] \psi(r, \theta, \phi) = E \psi(r, \theta, \phi). \quad (.46)$$

- (a) Comment on the physical significance of each term in this equation. Why can we factorize the wavefunction into a product of radial and angular functions $\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$? Why can we replace the $\hat{L}^2$ operator with $\ell(\ell+1)\hbar^2$?
- (b) Assuming the Coulomb potential $V(r) = -e^2/4\pi\epsilon_0 r$, explain the concept of an effective potential $V_{\text{eff}}(r)$ and what is the expression for this? What is the physical significance of the relative sign in front of the $\hat{L}^2$ and Coulomb terms?
- (c) Sketch the Coulomb potential, $\ell(\ell+1)\hbar^2/2m_e r^2$ piece, and effective $V_{\text{eff}}(r)$ for cases when the electron has (i) no angular momentum $\ell = 0$ and (ii) one unit of angular momentum $\ell = 1$. In the $\ell = 1$ case, explain the physical behaviour near and far from $r = 0$.
- (d) Change variables to $u(r) = rR(r)$ and show the radial equation becomes

$$\frac{d^2 u}{dr^2} - \frac{\ell(\ell+1)}{r^2} u + \frac{2}{r_B r} u = k^2 u. \quad (.47)$$

What are the expressions for k and r_B ? Explain their physical interpretation.

- (e) Show the derivations for the functional form of the radial wavefunction $R(r)$ in the asymptotic limits: (i) near $r/r_B \rightarrow 0$ and far (ii) $r/r_B \rightarrow \infty$ from the nucleus. You do not need to normalize the wavefunction.

HW10: Spin half and hydrogen (set Nov 24, due Dec 3)

- The spin operator is $\hat{S} = \hat{\sigma}/2$. Write down the three 2×2 Pauli matrices $\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z$.
 - Describe, in terms of operators and eigenvalues, the physical and mathematical similarities and differences between electron spin and orbital angular momentum.
 - By explicitly matrix multiplication, show that: (i) commutator $[\hat{S}_y, \hat{S}_z] = i\hbar\hat{S}_x$, and (ii) anticommutators $\{\hat{\sigma}_x, \hat{\sigma}_y\} = 0$ and $\{\hat{\sigma}_x, \hat{\sigma}_x\} = 2\hat{I}$, where $\hat{I}$ is the identity.
- A system of two Stern-Gerlach (SG) machines sees the first aligned along the z -axis then a second SG device aligned along $\mathbf{n} = (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)^T$:

$$\xrightarrow{\text{spin-half particles}} \boxed{\text{SG}_A} \xrightarrow{\text{transmits } |\uparrow\rangle \text{ along } \mathbf{z}} \boxed{\text{SG}_B} \xrightarrow{\text{transmits } |\lambda_+\rangle \text{ along } \mathbf{n}} \quad (.48)$$

- Electrons in state $|\chi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$ enter SG_A . Show that the expectation value of the force on the electrons is $\langle \hat{f}_z \rangle = (|\alpha|^2 - |\beta|^2)\mu_B \frac{\partial B_z}{\partial z}$, where $\mu_B = e\hbar/2m_e$.
 - An electron enters SG_B in a spin-up state $|\uparrow\rangle$. By solving the eigenvalue problem $\hat{\sigma} \cdot \mathbf{n} |\lambda_-, \mathbf{n}\rangle = \lambda_- |\lambda_-, \mathbf{n}\rangle$, show that the probability of measuring the spin-down state $|\lambda_-, \mathbf{n}\rangle$ after exiting SG_B is $P(\lambda_-, \mathbf{n} | \uparrow) = \sin^2(\frac{\theta}{2})$.
- Consider a system of two identical electrons.
 - Explain the concept of identical particles and the Pauli exclusion principle.
 - By applying the combined spin raising operator $\hat{S}_{+,12} = \hat{S}_{+,1} + \hat{S}_{+,2}$ onto the $|1, -1\rangle = |\downarrow\rangle_1 |\downarrow\rangle_2$ state, show that this gives $|1, 0\rangle = \frac{1}{\sqrt{2}} (|\downarrow\rangle_1 |\uparrow\rangle_2 + |\uparrow\rangle_1 |\downarrow\rangle_2)$. Explain why the $|1, 0\rangle$ state is considered entangled whereas $|1, -1\rangle$ is separable?
 - Describe the three quantum numbers and their physical interpretation that characterize the states of hydrogen gross structure. How are their allowed values?
 - Express the energies E_n in terms of the fine-structure constant α and electron mass m_e . What is the value of E_1 in eV and Joules? How does this change if we use the reduced mass $\mu = m_e m_p / (m_e + m_p)$ instead of a stationary proton?
 - What states can electrons occupy in the ground, first and second excited energies? What does *degeneracy* mean in this context? Explain your answers by counting states with the hydrogen quantum numbers and concept of electron (sub)shells.
 - What is meant by a hydrogenic atom? Write the expressions for the energy formula, ground and first excited state wavefunctions in terms of atomic number Z . How does the most probable radial electron position scale with Z ?